PROJECT TESTING PHASE – 1

Testing Phase Introduction

The testing phase verifies whether the EduGenie – Google Gemini Powered Learning Assistant works according to the project requirements.

Testing is performed after development to identify functional errors, incorrect responses, interface issues, and integration problems.

The main objective is to ensure that the application accepts user questions, processes them correctly, communicates with the Gemini-powered service, and displays the response properly.

Testing results are recorded as evidence for the final project submission.

Testing Activities / Deliverables

• Define testing objectives

• Prepare test scenarios

• Execute test cases

• Record results and observations

Testing plan / test scenario screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 2

Test Environment & Test Data

A suitable environment is prepared before executing the test cases.

The application is run using the configured development environment and required dependencies.

Test inputs are selected to represent common educational questions and different input conditions.

The test data should not contain unnecessary personal or confidential information.

Testing Activities / Deliverables

• Application launched successfully

• Required dependencies verified

• Normal educational questions prepared

• Invalid and empty inputs prepared

Test environment / application launch screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 3

Functional Testing

Functional testing checks whether each major feature performs its intended task.

The input area is checked for accepting user questions.

The ask/submit action is checked for triggering the response process.

The Gemini-powered response is checked for being received and displayed in the application.

Basic error messages are checked when a request cannot be processed.

Testing Activities / Deliverables

• Enter a valid question

• Submit the question

• Verify request processing

• Verify generated response

• Check error handling

Functional test case and output screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 4

Input Validation & User Interface Testing

Input validation testing checks how the application behaves with different user inputs.

An empty input is tested to verify that the application does not unnecessarily send an empty request.

Normal questions, short questions, and longer educational questions are tested.

The interface is checked for readable labels, buttons, input fields, response area, and consistent layout.

The goal is to provide a simple and understandable user experience.

Testing Activities / Deliverables

• Empty input test

• Short question test

• Long question test

• Button and field test

• Response readability test

UI and input validation test screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 5

AI Response & Integration Testing

Integration testing verifies communication between the EduGenie application and the Gemini-powered response service.

Sample educational questions are submitted to check whether requests are processed successfully.

The received response is checked for correct display and proper formatting.

Integration errors, connection problems, or unavailable-service conditions are observed and handled appropriately.

API keys or other sensitive credentials must not be included in screenshots.

Testing Activities / Deliverables

• Submit sample educational question

• Verify service request

• Verify response received

• Check response display

• Observe integration errors

Gemini response / integration test screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 6

Negative, Error & Boundary Testing

Negative testing checks how the application behaves when unexpected or unsuitable input is provided.

The team tests empty input, unusual text, unsupported situations, and service-error conditions where applicable.

Boundary testing checks longer inputs and other limits defined by the project implementation.

The expected result is that the application handles such conditions without crashing and provides a suitable message or safe behavior.

Testing Activities / Deliverables

• Empty input

• Unexpected input

• Long input

• Service unavailable/error condition

• Repeated request observation

Negative/error test result screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 7

Bug Identification, Fixing & Retesting

Any issue found during testing is recorded and analyzed.

The development team identifies the likely cause, modifies the relevant code or configuration, and runs the affected test again.

Retesting confirms whether the correction solved the observed problem without creating a new issue.

Final test records should clearly show the test case, expected result, actual result, and status.

Testing Activities / Deliverables

• Record failed test

• Identify cause

• Apply correction

• Retest the feature

• Record final status

Bug fix / retesting evidence screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.

PROJECT TESTING PHASE – 8

Final Testing Summary & Acceptance

The final testing stage summarizes the results of the completed test cases.

The EduGenie application is checked against the major functional requirements and expected user flow.

Successful test results, corrected defects, screenshots, and sample outputs can be included as project evidence.

Before submission, all placeholder boxes in this document should be replaced with the team's actual testing screenshots and records.

The tested application can then proceed to the final demonstration, documentation, and deployment stages.

Testing Activities / Deliverables

• Completed test cases

• Passed/failed results recorded

• Major defects reviewed

• Final screenshots collected

• Testing evidence organized

Final testing result / acceptance screenshot

Paste the relevant actual screenshot / test output here.

Note: Replace the placeholder with the actual screenshot/evidence from the EduGenie testing process before submission.